

Milestone II – Multi-agent Workflow

Mitigation Plan

Risks	Mitigation plan	
	Design and development of agent	Agent integration into process
Adversarial Attacks	Users might try to "jailbreak" the agent to get unauthorized financial advice.	Use a strict system instruction in Opal: <i>"You are a data analyst, not a financial advisor. Do not provide 'Buy/Sell' recommendations."</i>
Bias and Discrimination	The agent might over-prioritize recent "hype" over institutional positioning.	Diverse Sourcing: Program the "Market Data Analysis" asset to cross-reference news with hard data like institutional options flow
Misinformation and hallucinations	Gemini may misinterpret a complex options chain or a "fake news" headline.	Ensure the Opal workflow is strictly utilizing Google Search grounding to verify real-time data.
Data Privacy Violations	Storing sensitive analysis in Google Sheets without proper access control.	Share google sheets with specific users
Intellectual Property Risks	Dont Take stock direction from other users personal blogs, use geminis own interpretation of the data to make a prediction	Only take information from whitelisted sites: yahoo finance, bloomberg ect

Goal

The purpose of this project is to automate stock analysis using a multi-agent setup

Agent Roles Layout:

1. Data Collector Agent

- Role: Gather market data, news, and signals
- Input: Stock ticker
- Output: Raw market data + insights

2. Market Analyst Agent

- Role: Generate full market bias report
- Input: Raw data
- Output: Structured analysis (bullish/bearish, reasoning)

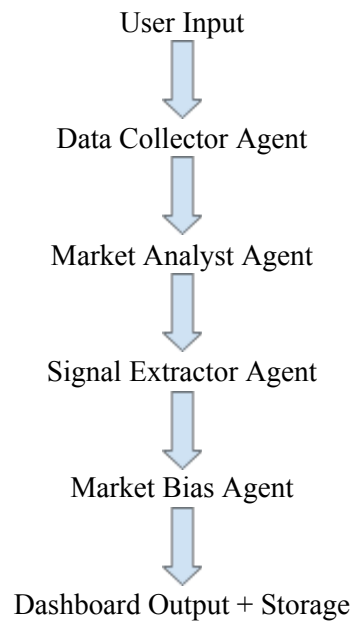
3. Signal Extractor Agent

- Role: Extract only high-conviction signals
- Input: Full report
- Output: Concise signals (spreadsheet-ready format)

4. Market Bias Report Agent

- Role: Generate HTML dashboard + store results
- Input: Analysis + signals
- Output: UI dashboard + Google Sheets entry

Data Flow



Coordination Strategy

Fixed Pipeline going through each agent

Error Handling

Fallback to previous version if incorrect execution, Logging the inputs in google sheets allow for the information to be logged and analyzed for validity

Tools

- **Gemini**
- **Google Opal**
- **Google Sheets**

